

Sampling design for optimal prediction in linear regression

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Abstract

Keywords:

1 Introduction

With the rapid growth of large-scale datasets, fitting statistical models using all available observations has become computationally prohibitive. Subsampling methods address this challenge by selecting an informative subset of data on which estimation is performed, thereby reducing computational cost while retaining statistical efficiency. A central question in this literature is: given a fixed budget of outcome observations, how should one design the sampling rule to minimize prediction error?

Existing approaches fall broadly into two categories. Information-based methods, exemplified by the Informative Based Optimal Subdata Selection (IBOSS) framework of Wang et al. (2019), select subdata deterministically by maximizing the Fisher information matrix, targeting covariate extremes to minimize parameter estimation variance. Probability-based importance sampling methods, by contrast, assign each observation a sampling probability and apply inverse probability weighting (IPW) to correct for the induced selection bias.

Lee et al. (2023) derived optimal sampling rules for binary-outcome electronic health record (EHR) data under a positive-only observation constraint, demonstrating that carefully designed sampling probabilities can substantially improve estimator efficiency relative to simple random sampling.

A common limitation of existing probability-based methods is the treatment of error variance. Most methods either assume homoscedasticity or focus on parameter estimation rather than prediction. In large-scale observational datasets, however, heteroscedasticity is the norm rather than the exception: regions of covariate space with higher noise are not uniformly more or less informative for prediction, and treating them as such leads to suboptimal sampling designs.

In this paper, we develop sampling rules that explicitly target prediction accuracy under heteroscedasticity. Working within a two-phase sampling framework, in which all covariates are observed but outcomes are expensive to collect, we derive the asymptotically optimal Bernoulli sampling probability for minimizing the excess prediction risk $R(\hat{\beta}_\pi) - R(\beta_0)$. The optimal probability takes the form

$$\pi_i^{\text{opt}} \propto \sigma(\mathbf{x}_i) \sqrt{\mathbf{x}_i^T A^{-1} \mathbf{x}_i}, \quad (1)$$

where $\sigma(\mathbf{x}_i)$ is the conditional standard deviation and $\mathbf{x}_i^T A^{-1} \mathbf{x}_i$ is the leverage score. Under homoscedasticity, this reduces to a square-root leverage rule, which is closely related to but distinct from standard leverage-proportional sampling.

Because $\sigma(\mathbf{x}_i)$ is unknown in practice, we propose a two-phase plugin procedure (OPT-hetero-plugin) that estimates the variance function from a small uniform pilot sample and uses the estimated scores to guide second-stage sampling. We compare this procedure against simple random sampling (UNI-IPW), leverage-proportional sampling (LEV-IPW), and the homoscedastic optimal rule (OPT-homo) through controlled simulation experiments and a large-scale real data application to NYC Yellow Taxi trip data ($N \approx 2.2$ million).

The simulation study reveals that the advantage of variance-aware sampling over variance-agnostic alternatives is governed not by the mere presence of heteroscedasticity, but by whether the high-variance region aligns with prediction-relevant directions in covariate space. When such alignment exists, OPT-hetero-plugin consistently reduces excess risk by 15–25% relative to OPT-homo across all subdata sizes considered, with the plugin estimator closely tracking oracle performance when the variance model is correctly specified. The real data application confirms that variance-aware sampling outperforms the homoscedastic rule when a meaningful variance driver can be identified, and performs comparably to leverage-based sampling when variance–leverage alignment is weak.

The remainder of the paper is organized as follows. Section 2 introduces the two-phase sampling framework, derives the asymptotic distribution of the IPW estimator, and establishes the optimal sampling probability. Section 3 presents the simulation study. Section 4 applies the methods to the NYC taxi dataset. Section 5 concludes with a discussion of limitations and future directions.

2 Proposed methods

2.1 Prediction model under two-phase sampling

Let y_i be a continuous outcome variable and $\mathbf{x}_i$ be a p -dimensional vector of covariates for the full data $i = 1, \dots, N$. We assume a standard linear regression model $Y = \mathbf{X}^T \boldsymbol{\beta} + \epsilon$, where $\mathbb{E}(\epsilon \mid \mathbf{X}) = 0$ and $\mathbb{V}(\epsilon \mid \mathbf{X}) = \sigma^2(\mathbf{X})$.

We consider a two-phase sampling design where the covariates $\mathbf{x}_i$ are cheap and observed for all N subjects, but the outcome y_i is expensive and observed only for a selected subsample. Let δ_i be the sampling indicator for the i -th subject, where $\delta_i = 1$ if y_i is observed and $\delta_i = 0$ otherwise. The sampling probability is denoted by $\pi(\mathbf{x}_i) = P(\delta_i = 1 \mid \mathbf{x}_i)$.

To estimate the regression coefficients $\boldsymbol{\beta}$, we minimize the inverse probability weighted

(IPW) squared error loss:

$$\mathcal{L}_N(\boldsymbol{\beta}) = \sum_{i=1}^N \frac{\delta_i}{\pi(\mathbf{x}_i)} (y_i - \mathbf{x}_i^T \boldsymbol{\beta})^2. \quad (2)$$

The estimator $\hat{\boldsymbol{\beta}}_\pi$ can be obtained in a closed form as:

$$\hat{\boldsymbol{\beta}}_\pi = \left(\sum_{i=1}^N \frac{\delta_i}{\pi(\mathbf{x}_i)} \mathbf{x}_i \mathbf{x}_i^T \right)^{-1} \sum_{i=1}^N \frac{\delta_i}{\pi(\mathbf{x}_i)} \mathbf{x}_i y_i. \quad (3)$$

The derivation follows by differentiating (2) with respect to $\boldsymbol{\beta}$ and setting the gradient to zero; see Supplementary S.1 for details.

We derive the asymptotic distribution of $\hat{\boldsymbol{\beta}}_\pi$ as follows.

Theorem 1 *Under regularity conditions, $N^{1/2}(\hat{\boldsymbol{\beta}}_\pi - \boldsymbol{\beta}_0) \rightarrow N(\mathbf{0}, \mathbf{A}^{-1} \mathbf{B}_\pi \mathbf{A}^{-1})$ in distribution, where $\boldsymbol{\beta}_0$ is the true parameter,*

$$\mathbf{A} = \mathbb{E}[\mathbf{X} \mathbf{X}^T], \quad \text{and} \quad \mathbf{B}_\pi = \mathbb{E} \left[\frac{\sigma^2(\mathbf{X})}{\pi(\mathbf{X})} \mathbf{X} \mathbf{X}^T \right].$$

The proof uses a Taylor expansion of the estimating equation, the law of large numbers, and the multivariate CLT; see Supplementary S.2 for details.

2.2 Optimal sampling for prediction

Our goal is to design the sampling probability $\pi(\cdot)$ such that the estimated model $\hat{\boldsymbol{\beta}}_\pi$ minimizes the Mean Squared Error (MSE) for future predictions. Let $(Y^*, \mathbf{X}^*)$ be a new, independent observation from the same population. The expected prediction MSE is:

$$R(\hat{\boldsymbol{\beta}}_\pi) = \mathbb{E} \left[\left(Y^* - \mathbf{X}^{*T} \hat{\boldsymbol{\beta}}_\pi \right)^2 \right].$$

Using the asymptotic distribution derived in Theorem 1, the leading order term of the

excess risk that depends on the sampling design π is given by:

$$\text{Excess Risk}(\pi) \approx \frac{1}{N} \text{tr}(\mathbf{A}^{-1} \mathbf{B}_\pi).$$

The derivation decomposes the prediction risk using the bias-variance structure and applies the asymptotic covariance from Theorem 1; see Supplementary S.3 for details.

Using the cyclic property of the trace operator, we can rewrite the core component as:

$$\text{tr}(\mathbf{A}^{-1} \mathbf{B}_\pi) = \mathbb{E} \left[\text{tr} \left\{ \mathbf{A}^{-1} \frac{\sigma^2(\mathbf{X})}{\pi(\mathbf{X})} \mathbf{X} \mathbf{X}^T \right\} \right] = \mathbb{E} \left[\frac{\sigma^2(\mathbf{X}) \mathbf{X}^T \mathbf{A}^{-1} \mathbf{X}}{\pi(\mathbf{X})} \right].$$

Based on the Cauchy-Schwarz inequality, minimizing this excess risk subject to a budget constraint $\sum_{i=1}^N \pi_i \leq n$ (where n is the target expected sample size) yields the optimal sampling score:

$$\Omega(\mathbf{X}) = \sigma^2(\mathbf{X}) \mathbf{X}^T \mathbf{A}^{-1} \mathbf{X}.$$

We consider two scenarios based on the error variance $\sigma^2(\mathbf{x})$.

Case 1: Homoscedastic Error

When the error variance is constant across the population, i.e., $\sigma^2(\mathbf{x}) \approx \sigma^2$, the term $\sigma(\mathbf{x})$ becomes a constant and cancels out. Because covariates $\mathbf{x}_i$ are fully observed for all N subjects, the matrix $\mathbf{A}$ can be empirically calculated without any pilot outcome data. The optimal sampling probability is:

$$\hat{\pi}_i^{\text{opt}} \propto \sqrt{\mathbf{x}_i^T \hat{\mathbf{A}}^{-1} \mathbf{x}_i}, \quad (4)$$

where $\hat{\mathbf{A}} = N^{-1} \sum_{i=1}^N \mathbf{x}_i \mathbf{x}_i^T$. This procedure is summarized in Algorithm 1.

Algorithm 1 MSE-Optimal Sampling (Homoscedastic)

Input: Covariate data $\{\mathbf{x}_i\}_{i=1}^N$, Target sample size n .

Step 1: Compute the empirical covariance matrix $\hat{\mathbf{A}} = N^{-1} \sum_{i=1}^N \mathbf{x}_i \mathbf{x}_i^T$.

Step 2: Calculate the leverage scores for each subject:

$$\Omega(\mathbf{x}_i) = \mathbf{x}_i^T \hat{\mathbf{A}}^{-1} \mathbf{x}_i$$

Step 3: Compute optimal probabilities $\hat{\pi}_i^{\text{opt}} \propto \sqrt{\Omega(\mathbf{x}_i)}$ satisfying $\sum_{i=1}^N \hat{\pi}_i^{\text{opt}} = n$.

Step 4: Sample $\delta_i \sim \text{Ber}(\hat{\pi}_i^{\text{opt}})$ for all $i = 1, \dots, N$, and observe y_i for selected subjects.

Step 5: Estimate final $\hat{\beta}_\pi$ by minimizing $\mathcal{L}_N(\beta)$ defined in (2).

Case 2: Heteroscedastic Error

In the general case where the error variance $\sigma^2(\mathbf{x})$ varies across the covariate space, we must account for regions with higher noise. To estimate $\sigma^2(\mathbf{x})$, a small uniform pilot sample is required. The optimal sampling probability is given by:

$$\hat{\pi}_i^{\text{opt}} \propto \hat{\sigma}(\mathbf{x}_i) \sqrt{\mathbf{x}_i^T \hat{\mathbf{A}}^{-1} \mathbf{x}_i}, \quad (5)$$

where $\hat{\sigma}(\mathbf{x}_i)$ is the estimated conditional standard deviation from the pilot model. This procedure is summarized in Algorithm 2.

Algorithm 2 MSE-Optimal Sampling (Heteroscedastic)

Input: Covariate data $\{\mathbf{x}_i\}_{i=1}^N$, Target sample size n , Pilot sample size n_0 .

Step 1: Compute the empirical covariance matrix $\hat{\mathbf{A}} = N^{-1} \sum_{i=1}^N \mathbf{x}_i \mathbf{x}_i^T$.

Step 2: Draw a uniform pilot sample of size n_0 and observe y_i . Estimate a pilot model $\tilde{\beta}$.

Step 3: Estimate conditional variance $\hat{\sigma}^2(\mathbf{x}_i)$ using squared residuals from the pilot model.

Step 4: Calculate scores incorporating both leverage and variance:

$$\Omega(\mathbf{x}_i) = \hat{\sigma}^2(\mathbf{x}_i) \left(\mathbf{x}_i^T \hat{\mathbf{A}}^{-1} \mathbf{x}_i \right)$$

Step 5: Compute optimal probabilities $\hat{\pi}_i^{\text{opt}} \propto \sqrt{\Omega(\mathbf{x}_i)}$ satisfying $\sum_{i=1}^N \hat{\pi}_i^{\text{opt}} = n$.

Step 6: Sample $\delta_i \sim \text{Ber}(\hat{\pi}_i^{\text{opt}})$ for all $i = 1, \dots, N$, and observe y_i for selected subjects.

Step 7: Estimate final $\hat{\beta}_\pi$ by minimizing $\mathcal{L}_N(\beta)$ defined in (2) using the sample selected in Step 6.

Theorem 2 *The sampling probability $\hat{\pi}_i^{\text{opt}}$ defined in (5) (or (4) under constant variance)*

minimizes the leading order term of the excess risk of the prediction MSE under the constraint $\sum_{i=1}^N \pi_i \leq n$.

The proof applies the Cauchy–Schwarz inequality to the optimization problem $\min_{\pi} \mathbb{E}[\Omega(\mathbf{X})/\pi(\mathbf{X})]$ subject to $\mathbb{E}[\pi(\mathbf{X})] \leq n/N$, yielding equality when $\pi(\mathbf{X}) \propto \sqrt{\Omega(\mathbf{X})}$; see Supplementary S.4 for details.

3 Simulation experiment

We evaluate the proposed sampling methods through a series of controlled simulation experiments. The experiments are designed to assess: (i) whether heteroscedasticity-aware sampling outperforms variance-agnostic alternatives in heteroscedastic settings; (ii) whether the plugin estimator approximates the oracle performance; and (iii) the sensitivity of the plugin method to pilot sample size and variance model misspecification.

3.1 Simulation Design

We fix the covariate structure and mean model across all experiments, varying only the error variance structure (homoscedastic vs. heteroscedastic).

3.1.1 Covariate Structure

We set $p = 9$ and define $\mathbf{x}_i = (1, \mathbf{Z}_i^T)^T$ where $\mathbf{Z}_i \in \mathbb{R}^9$. The covariates consist of three normally distributed, three centered binary, and three uniform variables, all generated independently:

$$\begin{aligned} Z_{i1}, Z_{i2}, Z_{i3} &\stackrel{\text{iid}}{\sim} N(0, 1), \\ Z_{i4}, Z_{i5}, Z_{i6} &\stackrel{\text{iid}}{\sim} \text{Bernoulli}(0.5) - 0.5, \quad Z_{ij} \in \{-0.5, 0.5\}, \\ Z_{i7}, Z_{i8}, Z_{i9} &\stackrel{\text{iid}}{\sim} \text{Uniform}(-1, 1). \end{aligned}$$

Binary variables are mean-centered to facilitate comparisons. The covariate structure is held fixed across all experiments.

3.1.2 Mean Model

The response is generated from a linear model $Y_i = \mathbf{x}_i^T \boldsymbol{\beta} + \epsilon_i$ with $\mathbb{E}(\epsilon_i \mid \mathbf{x}_i) = 0$. The regression coefficients are set to

$$\boldsymbol{\beta} = (1, 1.5, 1.0, 0.75, 1.0, 0.7, 0.5, 0.5, 0.25, 0)^T,$$

so that all three types of covariates contribute to the mean structure. The last component Z_{i9} is included as a noise variable with zero coefficient.

3.1.3 Error Variance Structure

Homoscedastic setting. The error term is drawn as $\epsilon_i \mid \mathbf{x}_i \sim N(0, 1)$. This setting is used as a sanity check to verify that heteroscedasticity-aware rules do not become unnecessarily unstable when the variance structure is flat.

Heteroscedastic setting. The error term is drawn as $\epsilon_i \mid \mathbf{x}_i \sim N(0, \sigma^2(\mathbf{Z}_i))$, where the variance function is

$$\sigma(\mathbf{Z}_i) = 0.5 + |0.5Z_{i1} + 0.5Z_{i2} + Z_{i6} + Z_{i8}|,$$

so that $\sigma^2(\mathbf{Z}_i) = [0.5 + |0.5Z_{i1} + 0.5Z_{i2} + Z_{i6} + Z_{i8}|]^2$. The variance function involves Z_{i1} and Z_{i2} , which have strong effects in the mean model, as well as Z_{i6} and Z_{i8} , which have weaker effects. This ensures that the variance structure only partially overlaps with the mean structure.

3.1.4 Conditional Monte Carlo Design

We fix a single full training dataset (seed = 123, $N = 10,000$) and an independent test dataset, then repeat only the subsampling procedure for each method across 500 replicates. The reported Monte Carlo Standard Error (MCSE) reflects the uncertainty due to subsampling randomness conditional on the fixed data, not the variability from repeated dataset generation. This design isolates differences between sampling rules by removing DGP-level variability. The FULL benchmark uses OLS on the entire training dataset and serves as a reference only; it is not a subsampling method.

3.2 Competing Methods and Performance Metrics

3.2.1 Competing Methods

We compare the following six methods:

1. **FULL**: OLS estimator using all N observations. Not a subsampling method; used as a performance reference only.
2. **SRS**: Simple random sampling with $\pi_i = k/N$, followed by IPW estimation.
3. **LEV-IPW**: Leverage-proportional sampling with $\pi_i \propto h_{ii}$, where $h_{ii} = \mathbf{x}_i^T \hat{\mathbf{A}}^{-1} \mathbf{x}_i$.
4. **OPT-homo**: Algorithm 1. The optimal rule under homoscedasticity.
5. **OPT-hetero-oracle**: Algorithm 2 using the true $\sigma(\mathbf{x}_i)$ directly. Serves as a theoretical performance upper bound.
6. **OPT-hetero-plugin**: Algorithm 2 with $\hat{\sigma}(\mathbf{x}_i)$ estimated from a pilot sample. A uniform pilot sample of size n_0 is drawn first to obtain $\tilde{\beta}_{\text{pilot}}$. The conditional standard deviation is then estimated by fitting

$$|\hat{e}_i| = \gamma_0 + \gamma_1 \cdot \text{abs_s}_i + \epsilon_i,$$

where $\text{abs_s}_i = |0.5Z_{i1} + 0.5Z_{i2} + Z_{i6} + Z_{i8}|$ is the absolute value of the variance driver, and γ_0, γ_1 are estimated by OLS. This two-parameter linear approximation assumes that the functional form of the variance driver is known. The final subsample of size k is then drawn using the estimated scores. This is a two-phase procedure; the additional pilot cost of n_0 outcome observations is explicitly controlled and examined in Section 3.3.3. In the main experiments (Sections 3.3.1 and 3.3.2), we assume the variance model is correctly specified. The case of misspecification is examined in Section 3.4.

All subsampling methods use Bernoulli sampling and normalize probabilities to satisfy $\sum_i \pi_i = k$. When $\pi_i > 1$, probabilities are clipped and redistributed uniformly.

3.2.2 Performance Metrics

The primary metric is the empirical excess prediction risk:

$$\widehat{\text{ER}}(\hat{\boldsymbol{\beta}}_\pi) = (\hat{\boldsymbol{\beta}}_\pi - \boldsymbol{\beta}_0)^\top \hat{\mathbf{A}} (\hat{\boldsymbol{\beta}}_\pi - \boldsymbol{\beta}_0),$$

where $\hat{\mathbf{A}} = N^{-1} \mathbf{X}^\top \mathbf{X}$ is the empirical second moment matrix of the covariates. This approximates the theoretical excess risk $\text{ER}(\boldsymbol{\beta}_\pi) = (\boldsymbol{\beta}_\pi - \boldsymbol{\beta}_0)^\top \mathbf{A} (\boldsymbol{\beta}_\pi - \boldsymbol{\beta}_0)$ by replacing $\mathbf{A} = \mathbb{E}[\mathbf{X}\mathbf{X}^\top]$ with $\hat{\mathbf{A}}$.

As supplementary metrics, we also report parameter MSE, $\frac{1}{p+1} \|\hat{\boldsymbol{\beta}}_\pi - \boldsymbol{\beta}_0\|^2$, and prediction MSE on the test set, $\frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} (\mu_i^* - \mathbf{x}_i^{*\top} \hat{\boldsymbol{\beta}}_\pi)^2$, where $\mu_i^* = \mathbf{x}_i^{*\top} \boldsymbol{\beta}_0$ is the true conditional mean.

Uncertainty across Monte Carlo replicates is summarized via MCSE. Method differences are assessed through paired differences; a 95% MCSE-based interval that excludes zero is taken as evidence of a consistent advantage.

3.3 Results

Results are presented in the order: homoscedastic sanity check, heteroscedastic main comparison, and pilot size sensitivity. All methods are compared under the same second-stage budget k . OPT-hetero-plugin involves an additional cost of n_0 pilot outcome observations relative to other methods; this additional cost is examined separately in Section 3.3.3.

3.3.1 Homoscedastic Sanity Check

This experiment verifies that the heteroscedasticity-aware rules do not exhibit unnecessary instability when the true error variance is constant. We set $N = 10,000$, $n_0 = 200, 500$ conditional replicates, and $k \in \{400, 600, 800, 1000, 2000\}$.

Table 1 reports excess risk for all methods. The oracle/homo and plugin/homo ratios were all within $\pm 2\%$ across all values of k , confirming that heteroscedasticity-aware rules operate stably in the homoscedastic case. As expected under this DGP—which uses a mixed covariate structure with a low proportion of extreme leverage points—leverage-based methods (LEV-IPW, OPT-homo) show little advantage over SRS.

Table 1: Excess risk (mean $\pm$ MCSE) under the homoscedastic setting. $N = 10,000$, $n_0 = 200, 500$ replicates.

Method	$k = 400$	$k = 600$	$k = 800$	$k = 1000$
FULL	0.00037	0.00037	0.00037	0.00037
SRS	$0.02499 \pm 5.29\text{e-}04$	$0.01625 \pm 3.24\text{e-}04$	$0.01199 \pm 2.53\text{e-}04$	$0.00954 \pm 1.98\text{e-}04$
LEV-IPW	$0.02605 \pm 5.52\text{e-}04$	$0.01640 \pm 3.46\text{e-}04$	$0.01204 \pm 2.52\text{e-}04$	$0.00954 \pm 1.98\text{e-}04$
OPT-homo	$0.02448 \pm 5.19\text{e-}04$	$0.01600 \pm 3.45\text{e-}04$	$0.01182 \pm 2.40\text{e-}04$	$0.00946 \pm 1.95\text{e-}04$
OPT-hetero-oracle	$0.02415 \pm 4.71\text{e-}04$	$0.01625 \pm 3.16\text{e-}04$	$0.01194 \pm 2.43\text{e-}04$	$0.00937 \pm 1.89\text{e-}04$
OPT-hetero-plugin	$0.02440 \pm 5.02\text{e-}04$	$0.01610 \pm 3.20\text{e-}04$	$0.01189 \pm 2.50\text{e-}04$	$0.00931 \pm 1.89\text{e-}04$

3.3.2 Main Heteroscedastic Comparison

The main experiment examines whether the hetero-aware gain is consistent across subdata sizes. Settings are identical to Section 3.3.1 except that the heteroscedastic DGP is used.

Table 2 reports excess risk and Table 3 reports paired differences for selected values of k . OPT-hetero-oracle and OPT-hetero-plugin achieved lower excess risk than SRS and OPT-homo across all values of k . The MCSE-based 95% intervals for SRS–plugin and homo–plugin were entirely positive, indicating a consistent gain within Monte Carlo uncertainty. The plugin/homo ratio ranged from 0.75 to 0.85, corresponding to a 15–25% reduction in excess risk. The absolute paired difference decreases as k grows, but the direction of the advantage is consistent throughout.

The plugin–oracle difference was small across all k , with MCSE-based intervals generally including zero, indicating that the plugin estimator closely approximates oracle performance when the variance model is correctly specified.

Table 2: Excess risk (mean $\pm$ MCSE) under the heteroscedastic setting. $N = 10,000$, $n_0 = 200, 500$ replicates.

Method	$k = 400$	$k = 600$	$k = 800$	$k = 1000$
FULL	0.00078	0.00078	0.00078	0.00078
SRS	$0.05814 \pm 1.33\text{e-}03$	$0.03778 \pm 8.45\text{e-}04$	$0.02813 \pm 6.55\text{e-}04$	$0.02233 \pm 4.88\text{e-}04$
LEV-IPW	$0.05636 \pm 1.30\text{e-}03$	$0.03501 \pm 8.01\text{e-}04$	$0.02598 \pm 5.76\text{e-}04$	$0.02038 \pm 4.50\text{e-}04$
OPT-homo	$0.05442 \pm 1.21\text{e-}03$	$0.03553 \pm 7.95\text{e-}04$	$0.02647 \pm 5.44\text{e-}04$	$0.02111 \pm 4.36\text{e-}04$
OPT-hetero-oracle	$0.04466 \pm 8.79\text{e-}04$	$0.03075 \pm 6.42\text{e-}04$	$0.02238 \pm 4.70\text{e-}04$	$0.01710 \pm 3.63\text{e-}04$
OPT-hetero-plugin	$0.04606 \pm 9.63\text{e-}04$	$0.03020 \pm 6.34\text{e-}04$	$0.02180 \pm 4.56\text{e-}04$	$0.01732 \pm 3.59\text{e-}04$

Table 3: Paired differences in excess risk (mean [approximate 95% MCSE interval]) under the heteroscedastic setting. Intervals are computed as mean $\pm 1.96 \times$ MCSE.

Contrast	$k = 400$	$k = 1000$	$k = 2000$
SRS – plugin	0.01208 [0.009, 0.015]	0.00501 [0.004, 0.006]	0.00310 [0.003, 0.004]
homo – plugin	0.00836 [0.005, 0.012]	0.00379 [0.003, 0.005]	0.00243 [0.002, 0.003]
plugin – oracle	0.00140 [−0.001, 0.004]	0.00022 [−0.001, 0.001]	−0.00017 [−0.001, 0.000]

3.3.3 Pilot Size Sensitivity

We examine the effect of pilot size n_0 on the plugin method’s performance when the variance model is correctly specified. We vary $n_0 \in \{50, 100, 200, 300, 400, 500\}$ with $k = 1000$ fixed and 500 conditional replicates.

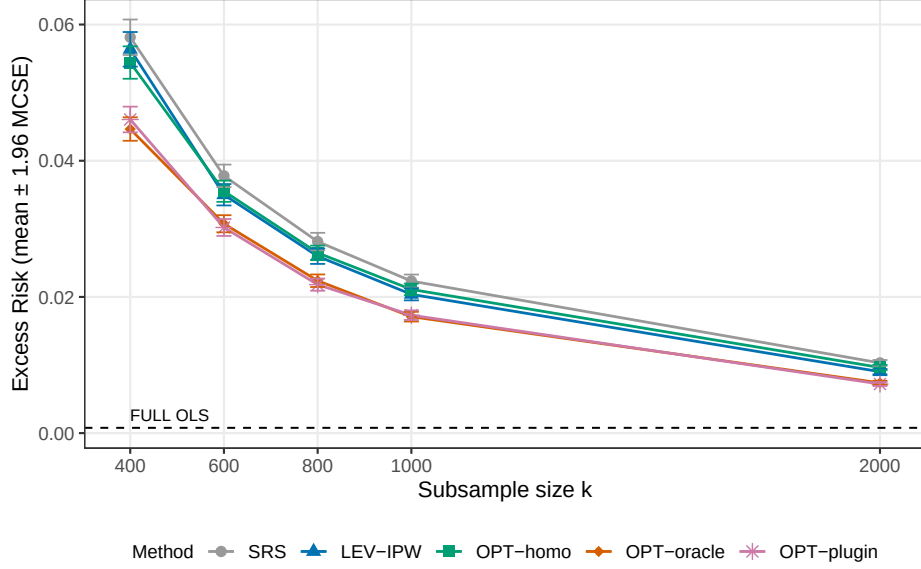


Figure 1: Excess risk by method and subdata size k under the heteroscedastic setting.

Table 4 reports plugin diagnostics and excess risk across pilot sizes. The sigma correlation and score correlation increased monotonically with n_0 (from 0.992 at $n_0 = 50$ to 0.9998 at $n_0 = 500$), yet excess risk was largely insensitive to n_0 . The plugin–oracle paired difference included zero in its MCSE-based interval for all n_0 , including $n_0 = 50$.

These results suggest that when the variance model is correctly specified and the variance signal is strong, even a small pilot sample is sufficient to recover the ranking of sampling scores. Under this setting, the choice of variance model specification appears to matter more than the pilot size. This point is further examined in Section 3.4.

Table 4: OPT-hetero-plugin performance by pilot size n_0 . $k = 1000$, 500 replicates.

n_0	Excess risk	MCSE	$\rho(\sigma, \hat{\sigma})$	$\rho(\Omega, \hat{\Omega})$
50	0.016644	0.000346	0.9924	0.9805
100	0.016039	0.000363	0.9985	0.9959
200	0.017321	0.000359	0.9993	0.9986
300	0.016828	0.000352	0.9996	0.9992
400	0.016659	0.000347	0.9997	0.9995
500	0.016473	0.000349	0.9998	0.9996

3.4 Robustness Check

We examine the sensitivity of OPT-hetero-plugin to variance model misspecification. We fix $N = 10,000$, $k = 1000$, $n_0 = 200$, and 500 replicates, and compare three misspecification levels:

Setting	Estimation model	Description
<code>correct_scale</code>	$ \hat{e}_i \sim \text{abs-s-full}$	Full variance driver
<code>mild_omitted</code>	$ \hat{e}_i \sim \text{abs-s-mild}$	Z_8 omitted
<code>severe_omitted</code>	$ \hat{e}_i \sim \text{abs-s-severe}$	Z_6 and Z_8 both omitted

Table 5 reports excess risk and Table 6 reports plugin diagnostics. Under `correct_scale`, the plugin–oracle interval included zero. From `mild_omitted` onward, the lower bound of the plugin–oracle interval became positive, and the gap was largest under `severe_omitted`. The pattern of decreasing sigma and score correlations with increasing misspecification is also visible in Figure 2.

Importantly, the SRS–plugin and homo–plugin lower bounds remained positive under both misspecified settings, indicating that OPT-hetero-plugin retains a meaningful advantage over variance-agnostic methods even when the variance model is partially misspecified. The gain is reduced relative to `correct_scale`, and we leave further investigation of variance model selection as a direction for future work.

Table 5: Excess risk by misspecification setting ($k = 1000$, $n_0 = 200$, 500 replicates).

Method	Mean ER	MCSE
FULL	0.000777	—
SRS	0.022333	0.000488
LEV-IPW	0.020382	0.000450
OPT-homo	0.021106	0.000436
OPT-hetero-oracle	0.017101	0.000363
plugin-correct	0.017321	0.000359
plugin-mild	0.018899	0.000386
plugin-severe	0.019752	0.000430

Table 6: Plugin diagnostics by misspecification setting.

Setting	$\rho(\sigma, \hat{\sigma})$	$\rho(\Omega, \hat{\Omega})$
correct_scale	0.9993	0.9986
mild_omitted	0.6699	0.7151
severe_omitted	0.4411	0.5813

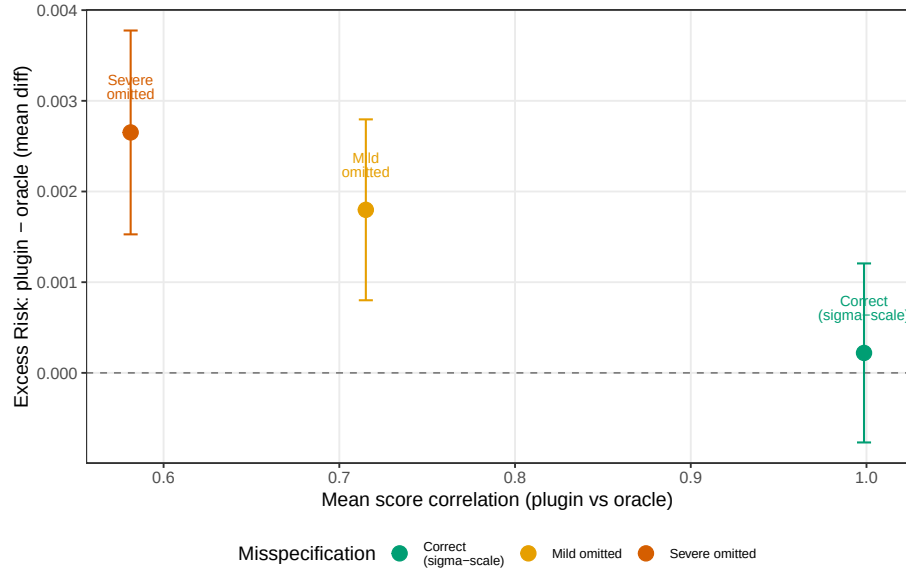


Figure 2: Score correlation vs. plugin-oracle excess risk gap across misspecification settings.

3.5 Computational Cost

We assess the runtime of the proposed method relative to simpler alternatives to confirm that the excess risk improvement comes at a reasonable computational cost.

We measure per-replicate runtime at $N = 100,000$ ($k = 1000$, 50 benchmarking replicates) with a staged decomposition, and additionally compare scaling across $N \in \{10,000, 50,000, 100,000, 500,000\}$ ($k = 1000$, 30 replicates).

Stage decomposition ($N = 100,000$). The total runtime of OPT-hetero-plugin was approximately 0.028 seconds, about 14 times that of SRS (0.002 seconds). The variance estimation stage accounted for 67.8% of the total cost and was the primary bottleneck. Pilot sampling and fitting together contributed less than 1.4%. Runtime was largely insensitive to k because variance estimation scales with N , not k (Table 7).

Table 7: Stage decomposition of OPT-hetero-plugin runtime at $N = 100,000$.

k	Pilot + variance (sec)	Total (sec)	Variance est. share
1000	0.0194 (69.2%)	0.0281	67.8%
3000	0.0204 (65.8%)	0.0309	64.7%
5000	0.0180 (61.9%)	0.0291	61.2%

Scaling with N . Table 8 reports runtime across methods and data sizes. The plugin/SRS ratio increased from approximately 10 at $N = 10,000$ to 32 at $N = 500,000$. At $N = 100,000$, the absolute runtime of 0.03 seconds per replicate is practically acceptable. For $N \geq 500,000$, efficiency improvements to the variance estimation stage may be warranted; we leave this as a direction for future work.

Table 8: Runtime (seconds per replicate) by method and data size N ($k = 1000$).

N	SRS	LEV-IPW	OPT-homo	OPT-plugin	plugin/SRS
10,000	0.0004	0.0012	0.0009	0.0039	9.8
50,000	0.0009	0.0053	0.0056	0.0147	16.3
100,000	0.0019	0.0104	0.0096	0.0296	15.6
500,000	0.0057	0.0709	0.0736	0.1838	32.2

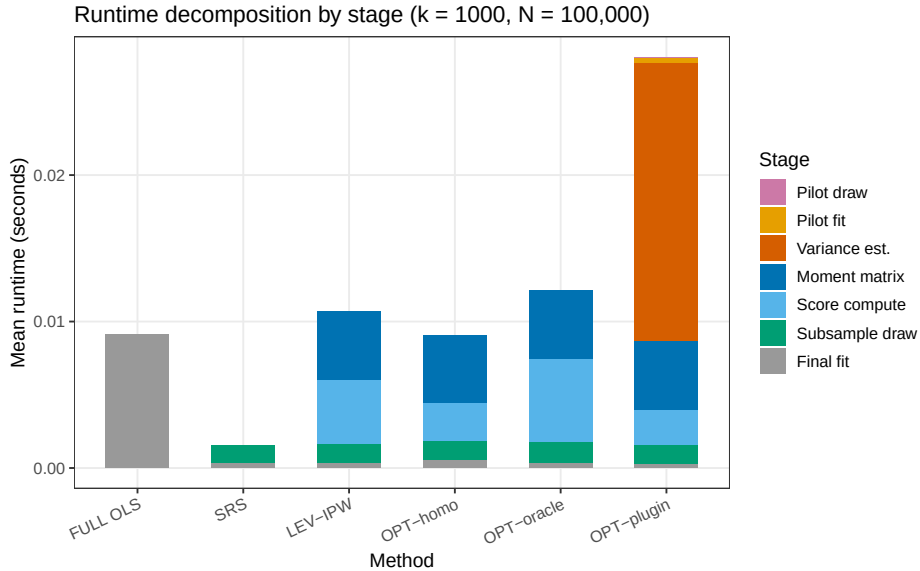


Figure 3: Stage decomposition of OPT-hetero-plugin runtime ($N = 100,000$, $k = 1000$).

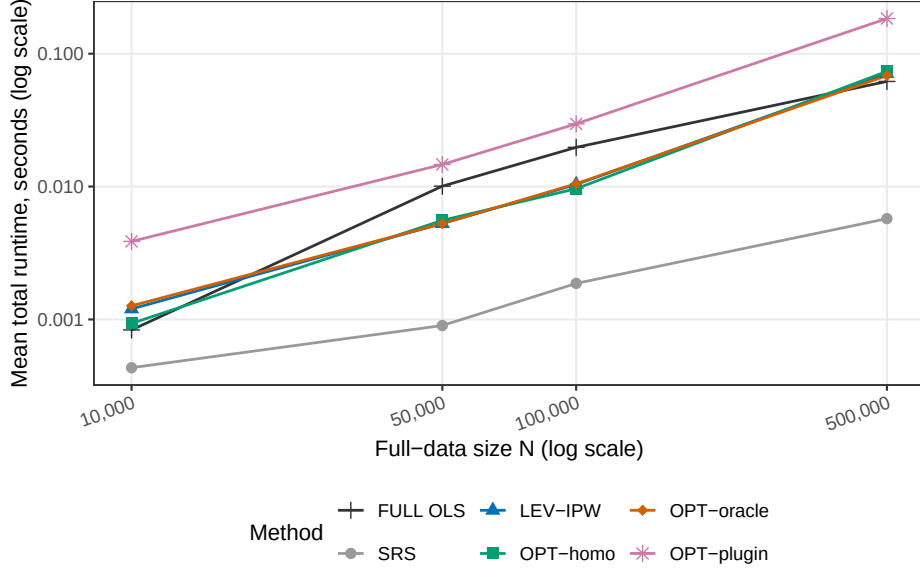


Figure 4: Runtime scaling with N across methods ($k = 1000$).

4 Real data application

We apply the proposed methods to a large-scale transportation dataset to assess their practical performance under real-world heteroscedasticity.

4.1 Data Description

We use the NYC Yellow Taxi Trip Data for January 2023, publicly available from the NYC Taxi and Limousine Commission (TLC). After applying the following preprocessing steps, the dataset contains $N = 2,737,913$ observations:

- Restricted to standard rate trips (`RatecodeID` = 1) to exclude flat-fare airport trips (e.g., JFK \$52 fixed fare) that violate linearity;
- Removed trips with `fare_amount` ≤ 2.5 or ≥ 200 , `trip_distance` ≤ 0 or ≥ 50 , or `trip_duration` ≤ 1 or ≥ 120 minutes.

The response variable is `fare_amount` (USD). The covariate vector is $\mathbf{x}_i = (1, \text{trip_distance}_i, \text{trip_duration}_i)$ giving $p = 4$.

Heteroscedasticity. A Breusch–Pagan test on a preliminary OLS fit yields $\text{BP} = 51,326$ ($\text{df} = 3, p < 2.2 \times 10^{-16}$), confirming strong heteroscedasticity. Residual plots show a fan-out pattern with increasing fitted values, motivating the use of variance-aware sampling.

4.2 Implementation Details

We split the data into a training set ($N = 2,190,330$, 80%) and a test set ($N_{\text{test}} = 547,583$, 20%). Following the two-phase sampling framework of Section 2, we treat all N covariate vectors as observed and treat the outcome `fare_amount` as observable only for the selected subsample. The FULL OLS estimator $\hat{\beta}_{\text{full}}$ on the training set serves as the pseudo-truth.

Variance model. To determine the variance driver for OPT-hetero-plugin, we fit $\log |\hat{e}_i| \sim \mathbf{x}_i$ on a preliminary pilot sample. Among the covariates, `trip_duration` is by far the dominant predictor ($t = 436.3$), which is intuitive: trips that take longer due to traffic or detours tend to have higher fare variability. We therefore use the model

$$|\hat{e}_i| = \gamma_0 + \gamma_1 \cdot \text{trip_duration}_i + \epsilon_i,$$

with $\hat{\sigma}(\mathbf{x}_i) = \max(\hat{\gamma}_0 + \hat{\gamma}_1 \cdot \text{trip_duration}_i, 10^{-6})$.

We compare the following methods: FULL, UNI-IPW, LEV-IPW, OPT-homo, and OPT-hetero-plugin. The pilot size is fixed at $n_0 = 500$, the second-stage budget is $k \in \{1000, 2500, 5000, 10000\}$, and all results are averaged over $B = 300$ conditional subsampling replicates.

4.3 Evaluation Metrics

The primary metric is the pseudo excess risk:

$$\widehat{\text{ER}}(\hat{\beta}_\pi) = (\hat{\beta}_\pi - \hat{\beta}_{\text{full}})^T \hat{\mathbf{A}} (\hat{\beta}_\pi - \hat{\beta}_{\text{full}}),$$

where $\hat{\beta}_{\text{full}}$ replaces the unknown β_0 as the pseudo-truth and $\hat{\mathbf{A}} = N^{-1}\mathbf{X}^T\mathbf{X}$. We additionally report test prediction MSE:

$$\widehat{\text{MSE}}_{\text{test}} = \frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} \left(y_i^* - \mathbf{x}_i^{*T} \hat{\beta}_{\pi} \right)^2.$$

4.4 Results

Tables 9 and 10 report pseudo excess risk and test MSE, respectively. Table 11 summarizes the paired differences.

Table 9: Pseudo excess risk (mean $\pm$ MCSE) on the NYC Yellow Taxi dataset. $N = 2,190,330$, $n_0 = 500$, 300 replicates.

Method	$k = 1000$	$k = 2500$	$k = 5000$	$k = 10000$
FULL	0	0	0	0
UNI-IPW	$0.02157 \pm 2.23\text{e-}03$	$0.01139 \pm 1.09\text{e-}03$	$0.00635 \pm 5.23\text{e-}04$	$0.00335 \pm 2.46\text{e-}04$
LEV-IPW	$0.00584 \pm 2.88\text{e-}04$	$0.00234 \pm 1.03\text{e-}04$	$0.00116 \pm 5.30\text{e-}05$	$0.00056 \pm 2.30\text{e-}05$
OPT-homo	$0.00983 \pm 5.30\text{e-}04$	$0.00373 \pm 2.07\text{e-}04$	$0.00192 \pm 1.01\text{e-}04$	$0.00089 \pm 4.50\text{e-}05$
OPT-hetero-plugin	$0.00561 \pm 2.68\text{e-}04$	$0.00209 \pm 8.20\text{e-}05$	$0.00107 \pm 4.60\text{e-}05$	$0.00054 \pm 2.50\text{e-}05$

Table 10: Test prediction MSE (mean $\pm$ MCSE) on the NYC Yellow Taxi dataset.

Method	$k = 1000$	$k = 2500$	$k = 5000$	$k = 10000$
FULL	1.1603	1.1603	1.1603	1.1603
UNI-IPW	$1.1819 \pm 2.20\text{e-}03$	$1.1717 \pm 1.07\text{e-}03$	$1.1667 \pm 5.09\text{e-}04$	$1.1637 \pm 2.37\text{e-}04$
LEV-IPW	$1.1662 \pm 2.91\text{e-}04$	$1.1626 \pm 1.04\text{e-}04$	$1.1615 \pm 5.30\text{e-}05$	$1.1609 \pm 2.60\text{e-}05$
OPT-homo	$1.1701 \pm 5.23\text{e-}04$	$1.1640 \pm 2.07\text{e-}04$	$1.1622 \pm 1.03\text{e-}04$	$1.1612 \pm 4.50\text{e-}05$
OPT-hetero-plugin	$1.1660 \pm 2.72\text{e-}04$	$1.1624 \pm 8.30\text{e-}05$	$1.1614 \pm 4.70\text{e-}05$	$1.1608 \pm 2.60\text{e-}05$

Table 11: Paired differences in pseudo excess risk. Significance based on 95% MCSE intervals.

Contrast	$k = 1000$	$k = 2500$	$k = 5000$	$k = 10000$
UNI-IPW – plugin	sig.	sig.	sig.	sig.
homo – plugin	sig.	sig.	sig.	sig.
LEV-IPW – plugin	n.s.	marginal	n.s.	n.s.
LEV-IPW – homo	n.s.	n.s.	n.s.	n.s.

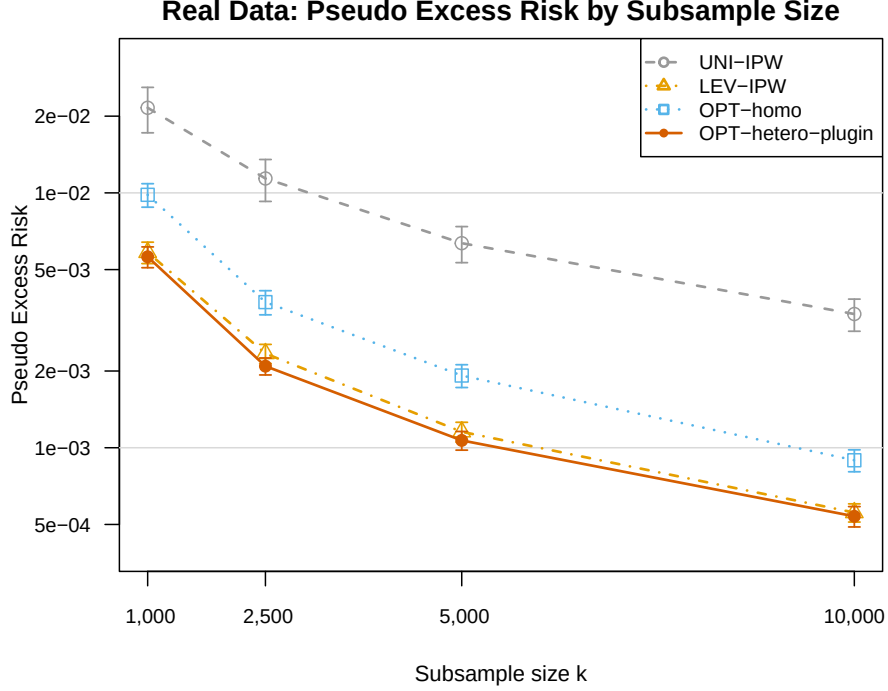


Figure 5: Pseudo excess risk by method and subdata size k on the NYC Yellow Taxi dataset.

OPT-hetero-plugin vs. OPT-homo. OPT-hetero-plugin achieves significantly lower pseudo excess risk than OPT-homo across all values of k . Specifying `trip_duration` as the variance driver provides a meaningful advantage over the homoscedastic assumption, consistent with the simulation results of Section 3.3.2.

OPT-hetero-plugin vs. LEV-IPW. The mean paired difference favors OPT-hetero-plugin at all k , but the 95% MCSE-based intervals include zero in most cases, indicating statistical equivalence between the two methods. This result can be attributed to the weak variance–leverage alignment in this dataset: EDA reveals a Spearman correlation of 0.177 between $\hat{\sigma}(\mathbf{x}_i)$ and h_{ii} . Because `trip_duration` contributes to both the mean structure and the variance, the plugin sampling score $\hat{\sigma}(\mathbf{x}_i)\sqrt{h_{ii}}$ generates rankings similar to those of the leverage score h_{ii} alone.

LEV-IPW vs. OPT-homo. Unlike in the simulation, LEV-IPW consistently shows lower pseudo excess risk than OPT-homo in the real data (mean difference negative throughout,

though non-significant). This reflects the flat leverage distribution of the dataset: the maximum leverage score is $h_{ii} = 0.006$ and the top 1% share only 8.3% of total leverage mass. With little concentration in high-leverage points, the square-root dampening of OPT-homo offers no advantage over the proportional weighting of LEV-IPW.

Summary. The overall ranking is $\text{OPT-hetero-plugin} \approx \text{LEV-IPW} > \text{OPT-homo} \gg \text{UNI-IPW}$. The real data results reproduce the key simulation finding that the plugin method outperforms the homoscedastic rule when the variance model is appropriately specified. The near-equivalence of plugin and LEV-IPW is consistent with the simulation regime in which variance–leverage alignment is weak, and extends the simulation findings to a realistic large-scale setting.

5 Discussion

5.1 Summary of findings

This paper proposed a prediction-optimal subsampling framework for large-scale linear regression under heteroscedasticity. The optimal sampling probability, derived by minimizing the leading term of the excess prediction risk via the Cauchy–Schwarz inequality, combines the leverage score with the conditional standard deviation. A practical two-phase plugin algorithm was developed and evaluated against variance-agnostic baselines in both simulation and real data settings.

The simulation study demonstrated three key findings. First, under correct variance model specification, OPT-hetero-plugin consistently achieves 15–25% lower excess risk than OPT-homo and matches oracle performance across all subdata sizes, confirming that the plugin procedure successfully captures the variance signal. Second, even under mild variance model misspecification (omission of one variance driver), OPT-hetero-plugin retains a statistically significant advantage over both UNI-IPW and OPT-homo, though the oracle

gap widens. Third, performance is largely insensitive to pilot size n_0 when the variance model is correctly specified: even $n_0 = 50$ suffices to recover the score ranking, suggesting that variance model specification matters more than pilot size in this setting.

The real data application on NYC Yellow Taxi data confirmed that OPT-hetero-plugin significantly outperforms OPT-homo when a meaningful variance driver (`trip_duration`) is identified. The near-equivalence of plugin and LEV-IPW in that setting is explained by the weak variance–leverage alignment (Spearman $\rho = 0.177$) and the flat leverage distribution, consistent with the simulation finding that the advantage of variance-aware sampling depends on the structural relationship between variance and covariate geometry.

5.2 Limitations

Variance model specification. The performance of OPT-hetero-plugin depends on correctly identifying the functional form of the variance driver. In the simulation, the variance driver was assumed known up to two parameters; in the real data, it was identified through a preliminary exploratory regression. In practice, selecting the variance model from the data without a principled procedure introduces a model selection step that is not currently accounted for in the theory. Extending the framework to accommodate nonparametric or data-adaptive variance estimation is a natural direction for future work.

Two-phase sampling cost. OPT-hetero-plugin incurs an additional n_0 outcome observations relative to single-phase methods. Although the total runtime at $N = 100,000$ is approximately 0.028 seconds per replicate (roughly 14 times that of UNI-IPW), the additional labeled-outcome cost may be a practical constraint in settings where outcome collection is expensive. The current experiments treat the pilot cost as separate from the second-stage budget; a budget-matched comparison (total $n_0 + k$ outcomes for plugin vs. k outcomes for other methods) remains to be studied.

Asymptotic optimality vs. finite-sample behavior. An important observation from the simulation experiments is that OPT-hetero-oracle, which uses the true $\sigma(\mathbf{x}_i)$ directly, does not always dominate OPT-homo in finite samples, even though it minimizes the asymptotic leading term. This instability is most pronounced when the variance function is strongly aligned with the leverage score, causing the oracle sampling probabilities to concentrate heavily in the tails. In such settings, the remainder term in the excess risk decomposition—which is not captured by the leading-term analysis—can offset the asymptotic advantage. This finding underscores that first-order asymptotic optimality does not automatically translate to finite-sample dominance, and that stabilization of extreme sampling weights may be necessary in practice.

Pilot sample exclusion from the final estimator. The current implementation uses the pilot sample only for variance estimation and draws an independent second-stage sample for the final IPW estimator. Incorporating the pilot observations into the final estimator (with appropriate weighting) could improve efficiency, particularly when n_0 is non-negligible relative to k . This design choice was left for future investigation.

5.3 Future directions

Distribution shift. The current framework assumes that the test distribution coincides with the training distribution, so that the prediction-risk matrix $A = E[\mathbf{X}\mathbf{X}^T]$ is common to both. In settings where the test covariate distribution differs from the training distribution—a common scenario in practice—the optimal sampling probability should account for the target distribution separately. Extending the theory to this distribution-shift setting would require redefining the excess risk objective with a separate test-distribution matrix and re-deriving the optimal π , and constitutes a natural direction for future work.

Data-adaptive variance model selection. The current plugin procedure assumes that the functional form of the variance driver is known. A fully data-driven approach that

selects or estimates the variance function without strong parametric assumptions—for example, using pilot residuals with a kernel smoother or a random forest—would broaden the applicability of the method.

Computational efficiency at very large scale. The variance estimation stage, which scales with N , accounts for approximately 60–70% of the total runtime. For $N \geq 500,000$, more efficient implementations (e.g., sketching-based approximations to the leverage scores, or batched variance estimation) may be warranted.

Connections to optimal experimental design. The derived optimal probability $\pi_i^{\text{opt}} \propto \sigma(\mathbf{x}_i) \sqrt{\mathbf{x}_i^T A^{-1} \mathbf{x}_i}$ has a natural interpretation in terms of classical optimal design theory: it is a prediction-optimal importance sampling rule that balances informativeness (leverage) with noise level (variance). Formalizing this connection and exploring whether analogous rules can be derived for other model classes (generalized linear models, survival models) is a promising direction.

References

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Supplementary material

S.1 Proof of the closed-form IPW estimator

Differentiating the IPW loss $\mathcal{L}_N(\boldsymbol{\beta}) = \sum_{i=1}^N \frac{\delta_i}{\pi(\mathbf{x}_i)} (y_i - \mathbf{x}_i^T \boldsymbol{\beta})^2$ with respect to $\boldsymbol{\beta}$ gives

$$\frac{\partial}{\partial \boldsymbol{\beta}} \mathcal{L}_N(\boldsymbol{\beta}) = -2 \sum_{i=1}^N \frac{\delta_i}{\pi(\mathbf{x}_i)} \mathbf{x}_i (y_i - \mathbf{x}_i^T \boldsymbol{\beta}).$$

Setting this to zero at $\boldsymbol{\beta} = \boldsymbol{\beta}^*$ and rearranging,

$$\left(\sum_{i=1}^N \frac{\delta_i}{\pi(\mathbf{x}_i)} \mathbf{x}_i \mathbf{x}_i^T \right) \boldsymbol{\beta}^* = \sum_{i=1}^N \frac{\delta_i}{\pi(\mathbf{x}_i)} \mathbf{x}_i y_i.$$

Assuming the left-hand matrix is invertible yields the closed-form estimator in (3).

S.2 Proof of Theorem 1

Estimating equation. Define

$$\psi_i(\boldsymbol{\beta}) := \frac{\delta_i}{\pi(\mathbf{x}_i)} (\mathbf{x}_i \mathbf{x}_i^T \boldsymbol{\beta} - \mathbf{x}_i y_i), \quad \Psi_N(\boldsymbol{\beta}) := \sum_{i=1}^N \psi_i(\boldsymbol{\beta}),$$

so that $\hat{\boldsymbol{\beta}}_\pi$ satisfies $\Psi_N(\hat{\boldsymbol{\beta}}_\pi) = 0$.

Taylor expansion. Since $\Psi_N(\boldsymbol{\beta})$ is linear in $\boldsymbol{\beta}$, the Taylor expansion around $\boldsymbol{\beta}_0$ has no remainder term:

$$0 = \frac{1}{\sqrt{N}} \Psi_N(\hat{\boldsymbol{\beta}}_\pi) = \frac{1}{\sqrt{N}} \Psi_N(\boldsymbol{\beta}_0) + \left\{ \frac{1}{N} \sum_{i=1}^N \frac{\delta_i}{\pi(\mathbf{x}_i)} \mathbf{x}_i \mathbf{x}_i^T \right\} \sqrt{N} (\hat{\boldsymbol{\beta}}_\pi - \boldsymbol{\beta}_0). \quad (\text{S.1})$$

Second term: law of large numbers. Define $W_i := \frac{\delta_i}{\pi(\mathbf{x}_i)} \mathbf{x}_i \mathbf{x}_i^T$. The LLN applies since X_i are iid and

$$\|X_i X_i^T\|_F = \sqrt{\sum_{j,k} (X_{ij} X_{ik})^2} < \infty \Rightarrow E[\|X_i\|^2] < \infty.$$

The unconditional expectation satisfies

$$E\left[\frac{\delta}{\pi(\mathbf{X})}\mathbf{X}\mathbf{X}^T\right] = E\left[\mathbf{X}\mathbf{X}^T \cdot E\left[\frac{\delta}{\pi(\mathbf{X})} \mid \mathbf{X}\right]\right] = E[\mathbf{X}\mathbf{X}^T] = \mathbf{A},$$

where the inner expectation equals 1 because $\pi(\mathbf{X}) = E[\delta \mid \mathbf{X}]$. By the LLN, $\frac{1}{N} \sum_{i=1}^N W_i \xrightarrow{p} \mathbf{A}$.

First term: central limit theorem. Writing $y_i = \mathbf{x}_i^T \boldsymbol{\beta}_0 + \varepsilon_i$,

$$\frac{1}{\sqrt{N}} \Psi_N(\boldsymbol{\beta}_0) = -\frac{1}{\sqrt{N}} \sum_{i=1}^N Z_i, \quad Z_i := \frac{\delta_i}{\pi(\mathbf{x}_i)} \mathbf{x}_i \varepsilon_i.$$

The multivariate CLT applies since:

1. Z_i are iid, as $(X_i, \varepsilon_i, \delta_i)$ are iid;
2. $E[Z_i] = 0$, since $E[\delta/\pi(\mathbf{X}) \cdot \varepsilon \mid \mathbf{X}] = E[\varepsilon \mid \mathbf{X}] = 0$;
3. $E[\|Z_i\|^2] < \infty$: noting $\delta_i^2 = \delta_i$ and $E[\delta_i \mid \mathbf{x}_i] = \pi(\mathbf{x}_i)$,

$$E[\|Z_i\|^2] = E\left[\frac{\sigma^2(\mathbf{X})}{\pi(\mathbf{X})} \|\mathbf{X}\|^2\right] < \infty.$$

The covariance matrix of Z_i is

$$\text{Var}(Z_i) = E[Z_i Z_i^T] = E\left[\frac{\delta_i^2}{\pi(\mathbf{x}_i)^2} \mathbf{x}_i \varepsilon_i \varepsilon_i^T \mathbf{x}_i^T\right] = E\left[\frac{\sigma^2(\mathbf{X})}{\pi(\mathbf{X})} \mathbf{X} \mathbf{X}^T\right] = \mathbf{B}_\pi,$$

where we used $\delta_i^2 = \delta_i$, $E[\delta_i \mid \mathbf{x}_i] = \pi(\mathbf{x}_i)$, and $E[\varepsilon_i^2 \mid \mathbf{x}_i] = \sigma^2(\mathbf{x}_i)$. By the multivariate CLT, $\frac{1}{\sqrt{N}} \sum_{i=1}^N Z_i \xrightarrow{d} N(\mathbf{0}, \mathbf{B}_\pi)$.

Conclusion. Substituting into (S.1),

$$\mathbf{A} \sqrt{N}(\hat{\boldsymbol{\beta}}_\pi - \boldsymbol{\beta}_0) = \frac{1}{\sqrt{N}} \sum_{i=1}^N Z_i + o_p(1).$$

Applying Slutsky's theorem,

$$\sqrt{N}(\hat{\beta}_\pi - \beta_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{A}^{-1}\mathbf{B}_\pi\mathbf{A}^{-1}).$$

S.3 Derivation of the excess risk approximation

Let $(Y^*, \mathbf{X}^*)$ be a new observation from the same population, with $Y^* = \mathbf{X}^{*\top}\beta_0 + \varepsilon^*$ and $E[\varepsilon^* | \mathbf{X}^*] = 0$. Expanding the prediction risk,

$$R(\hat{\beta}_\pi) = E[(Y^* - \mathbf{X}^{*\top}\hat{\beta}_\pi)^2] = E[(Y^* - \mathbf{X}^{*\top}\beta_0)^2] + E[(\mathbf{X}^{*\top}(\beta_0 - \hat{\beta}_\pi))^2] + 2E[(Y^* - \mathbf{X}^{*\top}\beta_0)\mathbf{X}^{*\top}(\beta_0 - \hat{\beta}_\pi)]$$

Cross term is zero. Since β_0 is the true parameter, (X^*, ε^*) is drawn from the same population as but independently of the training data, and $\hat{\beta}_\pi$ is a function of the training data only,

$$\begin{aligned} E[\varepsilon^*\mathbf{X}^{*\top}(\beta_0 - \hat{\beta}_\pi)] &= E\left[E[\varepsilon^*\mathbf{X}^{*\top}(\beta_0 - \hat{\beta}_\pi) | \mathbf{X}^*, \hat{\beta}_\pi]\right] \\ &= E[\mathbf{X}^{*\top}(\beta_0 - \hat{\beta}_\pi) \cdot E[\varepsilon^* | \mathbf{X}^*]] = 0. \end{aligned}$$

Excess risk. Let $u = \hat{\beta}_\pi - \beta_0$. Since u and $\mathbf{X}^*$ are independent and $E[\mathbf{X}^*\mathbf{X}^{*\top}] = \mathbf{A}$,

$$R(\hat{\beta}_\pi) - R(\beta_0) = E[u^\top \mathbf{X}^* \mathbf{X}^{*\top} u] = E[u^\top \mathbf{A} u] = \text{tr}(\mathbf{A} E[uu^\top]).$$

From Theorem 1, $\sqrt{N}u \xrightarrow{d} N(\mathbf{0}, \mathbf{A}^{-1}\mathbf{B}_\pi\mathbf{A}^{-1})$, so $E[uu^\top] \approx \frac{1}{N}\mathbf{A}^{-1}\mathbf{B}_\pi\mathbf{A}^{-1}$. Therefore

$$R(\hat{\beta}_\pi) - R(\beta_0) \approx \frac{1}{N}\text{tr}(\mathbf{A} \cdot \mathbf{A}^{-1}\mathbf{B}_\pi\mathbf{A}^{-1}) = \frac{1}{N}\text{tr}(\mathbf{B}_\pi\mathbf{A}^{-1}) = \frac{1}{N}\text{tr}(\mathbf{A}^{-1}\mathbf{B}_\pi),$$

where the last equality uses the cyclic property of the trace.

S.4 Proof of Theorem 2

Reduction to an optimization problem. From Supplementary S.3, minimizing the excess risk is equivalent to minimizing $\text{tr}(\mathbf{A}_\pi^{-1})$. Using the cyclic property and linearity of trace and expectation,

$$\text{tr}(\mathbf{B}_\pi \mathbf{A}^{-1}) = E \left[\frac{\sigma^2(\mathbf{X})}{\pi(\mathbf{X})} \text{tr}(\mathbf{X} \mathbf{X}^T \mathbf{A}^{-1}) \right] = E \left[\frac{\sigma^2(\mathbf{X})}{\pi(\mathbf{X})} \mathbf{X}^T \mathbf{A}^{-1} \mathbf{X} \right] = E \left[\frac{\Omega(\mathbf{X})}{\pi(\mathbf{X})} \right],$$

where $\Omega(\mathbf{X}) = \sigma^2(\mathbf{X}) \mathbf{X}^T \mathbf{A}^{-1} \mathbf{X}$. The optimization problem is therefore

$$\min_{\pi} E \left[\frac{\Omega(\mathbf{X})}{\pi(\mathbf{X})} \right] \quad \text{subject to} \quad E[\pi(\mathbf{X})] \leq \frac{n}{N}.$$

Optimal solution via Cauchy–Schwarz. Applying the Cauchy–Schwarz inequality,

$$E \left[\frac{\Omega(\mathbf{X})}{\pi(\mathbf{X})} \right] \cdot E[\pi(\mathbf{X})] \geq \left(E \left[\sqrt{\Omega(\mathbf{X})} \right] \right)^2,$$

with equality if and only if $\pi(\mathbf{X}) \propto \sqrt{\Omega(\mathbf{X})}$. The optimal probability is therefore

$$\pi_i^{\text{opt}} = \frac{\sqrt{\Omega(\mathbf{x}_i)}}{\sum_{j=1}^N \sqrt{\Omega(\mathbf{x}_j)}} \cdot n.$$

$C_\pi \geq 0$ and asymptotic improvement. Define $C_\pi := \text{tr}(\mathbf{B}_\pi \mathbf{A}^{-1}) - \text{tr}(\mathbf{B}_{\pi^{\text{opt}}} \mathbf{A}^{-1}) \geq 0$.

Since

$$N(R(\hat{\boldsymbol{\beta}}_\pi) - R(\boldsymbol{\beta}_0)) \xrightarrow{p} \text{tr}(\mathbf{B}_\pi \mathbf{A}^{-1}), \quad N(R(\hat{\boldsymbol{\beta}}_{\pi^{\text{opt}}}) - R(\boldsymbol{\beta}_0)) \xrightarrow{p} \text{tr}(\mathbf{B}_{\pi^{\text{opt}}} \mathbf{A}^{-1}),$$

it follows that $N(R(\hat{\boldsymbol{\beta}}_\pi) - R(\hat{\boldsymbol{\beta}}_{\pi^{\text{opt}}})) \xrightarrow{p} C_\pi \geq 0$, confirming that π^{opt} minimizes the leading term of the prediction MSE.

Plugin estimator: asymptotic equivalence. In practice, $\mathbf{A}$ and $\sigma^2(\mathbf{x}_i)$ are replaced by pilot estimates $\hat{\mathbf{A}}$ and $\hat{\sigma}^2(\mathbf{x}_i)$:

$$\hat{\pi}_i^{\text{opt}} = n \times \frac{\sqrt{\hat{\sigma}^2(\mathbf{x}_i) \mathbf{x}_i^T \hat{\mathbf{A}}^{-1} \mathbf{x}_i}}{\sum_{j=1}^N \sqrt{\hat{\sigma}^2(\mathbf{x}_j) \mathbf{x}_j^T \hat{\mathbf{A}}^{-1} \mathbf{x}_j}}.$$

Under the following regularity conditions:

- (R1) X_i are iid with $E[\|X_i X_i^T\|] < \infty$, so that $\hat{\mathbf{A}} \xrightarrow{p} \mathbf{A}$ by the LLN;
- (R2) A is invertible, so $P(\det(\hat{\mathbf{A}}) \neq 0) \rightarrow 1$ and $\hat{\mathbf{A}}^{-1} \xrightarrow{p} \mathbf{A}^{-1}$ by CMT;
- (R3) The pilot-based variance estimator is consistent: $\hat{\sigma}(\mathbf{x}_i) \xrightarrow{p} \sigma(\mathbf{x}_i)$.

Since the numerator and denominator of $\hat{\pi}_i^{\text{opt}}$ are continuous functions of $(\hat{\mathbf{A}}^{-1}, \hat{\sigma}(\mathbf{x}_i))$, the continuous mapping theorem gives

$$\hat{\pi}_i^{\text{opt}} \xrightarrow{p} \pi_i^{\text{opt}}.$$

To show $\text{tr}(B_{\hat{\pi}^{\text{opt}}} \hat{\mathbf{A}}^{-1}) \xrightarrow{p} \text{tr}(B_{\pi^{\text{opt}}} \mathbf{A}^{-1})$, note that by the LLN,

$$\frac{1}{N} \sum_{i=1}^N \frac{\sigma^2(\mathbf{x}_i)}{\hat{\pi}_i^{\text{opt}}} \mathbf{x}_i^T \hat{\mathbf{A}}^{-1} \mathbf{x}_i$$

converges in probability termwise: for each i , the map $g(a, B) = \frac{\sigma^2(\mathbf{x}_i)}{a} \mathbf{x}_i^T B \mathbf{x}_i$ is continuous for $a > 0$, so CMT applied to $(\hat{\pi}_i^{\text{opt}}, \hat{\mathbf{A}}^{-1}) \xrightarrow{p} (\pi_i^{\text{opt}}, \mathbf{A}^{-1})$ gives convergence of each term. Under the stated consistency and suitable domination conditions, the empirical average converges to the oracle counterpart, and since $\text{tr}(\cdot)$ is continuous, a final application of CMT yields

$$\text{tr}(B_{\hat{\pi}^{\text{opt}}} \hat{\mathbf{A}}^{-1}) \xrightarrow{p} \text{tr}(B_{\pi^{\text{opt}}} \mathbf{A}^{-1}).$$

Therefore

$$C_{\hat{\pi}} = \text{tr}(B_{\hat{\pi}^{\text{opt}}} \hat{\mathbf{A}}^{-1}) - \text{tr}(B_{\pi^{\text{opt}}} \mathbf{A}^{-1}) \xrightarrow{p} 0,$$

and $\hat{\pi}^{\text{opt}}$ also minimizes the leading term of the prediction MSE.